Forecasting Next-Week Volatility of S&P 500 Stocks: Revised Proposal

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1. Project question

For each S&P 500 stock and week, we forecast next week's log realized variance, measured by the Garman–Klass estimator from daily prices [4]. Models train on 2005–2015, tune on 2016–2018 and are tested on 2019–2026 with yearly expanding refits, a one-week gap at each boundary and no tuning on test data.

1. **Accuracy.** Do pooled models fit on all stocks (ridge and lasso, then gradient-boosted trees) beat a HAR regression fit per stock [2] on mean QLIKE [6] and MSE? The Diebold–Mariano test [3] runs at 5% on the weekly cross-sectional mean loss difference with Newey–West errors. Per-stock HAR needs 52 weeks of history (younger stocks use a pooled HAR), and every model is scored on the same rows.
2. **Attribution and regimes.** Which feature blocks carry any gain (refitting without each block), and does the gain keep its sign in calm weeks and in stress weeks (VIX ≥ 30 on the Friday)?

It is answered when, for each model, the test-period table of losses and Diebold–Mariano statistics against HAR, and the block-ablation table split by regime, exist.

How and why it changed. The economic-value question (volatility-targeted portfolio, Sharpe ratio, drawdown, Fama–French regression) is dropped, as the staff notes require. Earnings timing will come from the filing times of SEC 8-K Item 2.02 releases, a proxy for the announcement, not quarterly filings that arrive weeks after it. Fundamentals from XBRL are deferred: they start in mid-2009 and the question does not need them. Stress is defined by VIX at prediction time, not by calendar years, and qlib's Alpha158 price-volume factors [7] join the features.

2. Data

S&P 500 membership by date (2,720 snapshots, 977 tickers since 2005, 950 companies once renames are merged); daily prices from Yahoo Finance (677 tickers, 2005–2026) and Alpaca (from 2016, for 165 members Yahoo lacks, plus 5-minute bars); market state from FRED (VIX, 3-month VIX, credit spread, curve slope, T-bill); earnings-release filing times and industry codes from SEC EDGAR (code ready, fetch pending). The raw data is 94 MB and 3,423,011 stock-days, 2,315,615 after cleaning. The modeling table has **473,684 rows and 180 columns**: one row per stock and week (725 companies, 1,139 weeks), holding the target and every feature known at that Friday's close.

3. What we learned from the data

- **Survivorship.** Yahoo drops delisted companies. We recovered 30 renamed tickers (BK became BNY) and filled members Yahoo lacks from Alpaca from 2016; 108 tickers, all gone before 2016, have no free data. Coverage of index members is 55% in 2005, 99.4% in 2019 and 99.8% in 2026, so pre-2016 training data leans toward survivors. CRSP through WRDS would close the gap; we will ask the TF.
- **Ticker reuse and wrong series.** Yahoo reuses symbols (today's `FB` is an ETF), so prices are kept only near each ticker's index spells; 9 series inside a spell belong to other securities (Yahoo's `PARA` trades near \$100,000), and 4 change company on a date (Yahoo's `JCI` before 2016 is Tyco); hand-checked tables remove both.
- **The target is a sound proxy.** Against 5-minute realized variance for 24 current members since 2016 (12,515 stock-weeks), weekly Garman–Klass variance has correlation 0.95, easing the proposal's worry about noisy daily targets; delisted and pre-2016 stocks are not covered by this check.
- **Stress is rare.** 7.3% of stock-weeks are stress weeks, so results must be reported per regime, not as one pooled loss.
- **Leakage.** Feature windows end at the Friday close (unit tests truncate the data and recompute), an 8-K filed after 16:00 counts for the next week, membership is point-in-time (the last snapshot, 2026-08-18, is carried forward), and every price feature is a ratio, so back-adjusted prices cannot leak.

4. Plan for the rest of the project

Features: HAR lags (day, week, four weeks); Parkinson, Rogers–Satchell, overnight and downside variance; returns; dollar volume and Amihud illiquidity; SPY variance; VIX, 3-month VIX, credit spread, curve slope, T-bill; Alpha158; next, weeks since earnings, an expected-earnings flag and industry. **Models**: per-stock HAR and GARCH(1,1) baselines; pooled ridge and lasso on standardized features with training-window imputation; gradient-boosted trees (LightGBM). Clustering and neural networks only if time remains [1, 5]. **Evaluation**: QLIKE and MSE on 2019–2026, Diebold–Mariano tests against HAR, block ablation, every result split into calm and stress weeks.

5. Staff notes and the AC209A requirement

Economic-value backtest: dropped. Regularized regression and boosted trees: the two pooled models of Q1; clustering and neural networks wait. yfinance failures and missing delisted stocks: fetches are cached and resumable, renames, wrong series and the Alpaca fill are documented, and coverage is reported. XBRL only from mid-2009: fundamentals deferred. AC209A outside method: HAR, GARCH(1,1), the QLIKE loss and the Diebold–Mariano test, none covered in class.

Use of generative AI. Claude Code (Anthropic) helped write the data pipeline, tests and this text; the group reviewed every step and verified the numbers, which the notebook computes and exports.

References

[1] Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023.

[2] Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2): 174–196, 2009. doi: 10.1093/jjfinec/nbp001.

[3] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13 (3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

[4] Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980. doi: 10.1086/296072.

[5] Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020. doi: 10.1093/rfs/hhaa009.

[6] Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

[7] Xiao Yang, Weiqing Liu, Dong Zhou, Jiang Bian, and Tie-Yan Liu. Qlib: An AI-oriented quantitative investment platform, 2020.